

Lab equipment and techniques

A GUIDE TO THE BASIC APPARATUS IN A CHEMISTRY LAB AND HOW IT IS USED.

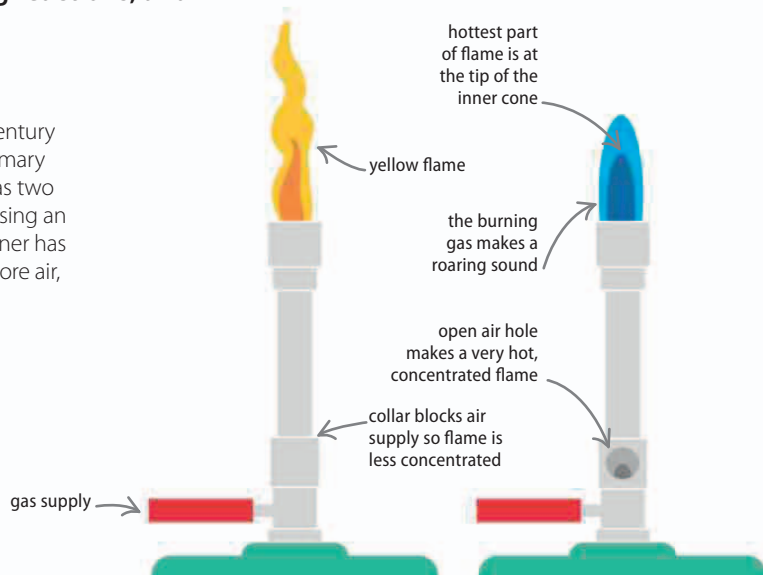
Every chemistry lab has some basic apparatus that can be used for heating substances, observing reactions, and finding out more about materials.

Bunsen burner

This simple gas burner was designed in the late 19th century by German chemist Robert Bunsen. It is used as the primary source of heat in chemistry experiments. The burner has two main settings that can be adjusted by opening and closing an air hole on its base. When the air hole is closed, the burner has a luminous flame. When the hole is opened, it lets in more air, which creates a very hot and roaring blue flame.

▷ Different flames

The roaring blue flame is used to heat reactants and boil liquids during experiments. The luminous flame, which is taller and not so hot, is used to ignite splints and burn powders in flame tests (see page 130).

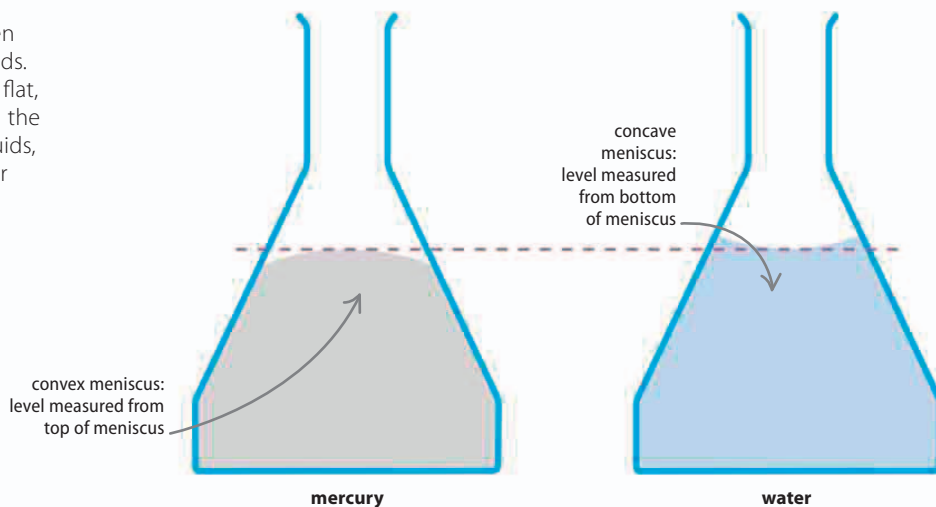


Measuring liquids

Chemists must be careful when measuring the volume of liquids. The surfaces of liquids are not flat, but have a curved edge called the meniscus. Water, like most liquids, has a concave meniscus. Other liquids, such as mercury, have a convex surface.

▷ Measure at eye level

To measure a volume, the eye should be level with the meniscus.



SEE ALSO

- ◀ 98–99 States of matter
- ◀ 106–107 Separating mixtures
- ◀ 128–129 Chemical reactions
- ◀ 130–131 Combustion